

The limitations of attentional resources across developmental groups: A three-dimensional multiple object tracking study

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Abstract

This study investigated the resource limits for dynamic visual attention across development using a three-dimensional multiple object-tracking (3D-MOT) task. Previous studies exploring such attentional limits were restricted by (i) the assessment of MOT in two dimensions, and (ii) categorical speed manipulations. 3D-MOT was used to measure attentional processing capacity when manipulating the number of target items and trial speed. Twenty-one participants grouped by age: school-aged (6-12 years), adolescent (13-18), or adult (19-30) group; were asked to follow 1 through 4 target spheres (separate conditions) among 8 total spheres, which moved randomly in a virtual volumetric space during an 8 second trial. The speed of the spheres changed for subsequent trials until threshold performance was attained, defined as the average speed at which *all* target spheres for each condition (1 through 4) could be tracked. All participants were able to track up to four items successfully; however, school-aged children were limited to slower trial speeds compared to the adolescent and adult groups ($p < .001$) across all conditions (i.e., 1 through 4 items). The decreasing trend in speed threshold scores at increasing levels of target items resembled a logarithmic function for all age groups ($r^2 = .983$, $p = .008$). This finding describes the deployment of attentional resources, where an upper limit of object capacity (when speed is close to motionless) was 6.48 for adults, 7.12 for adolescents, and 5.42 for school-aged, differing from limits described by two-dimensional paradigms (Alvarez & Franconeri, 2007). The decreasing trend is representative of the *flexible-resource model*, where in this case resource allocation is a function of trial speed and number of target items. These preliminary findings suggest that 3D-MOT can be used for characterizing the development of resource allocation in attentional processes through the use of a measure that best approximate real-world conditions.

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